Central Limit Theorem

ECON 97: Economic Toolkit

Patrick Molligo

UCLA

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iClicker course link

https://join.iclicker.com/RDNG

Quiz: continuous variables

Suppose $Z \sim N(0, 1)$. What is

$$P(-1.2 < Z < 0.8)?$$

A. 0.327

B. 0.673

C. 0.6738

D. 0.9032

Normal distribution practice

If $X \sim N(7, 25)$, find:

$$P(|X - 7| \leq 10)$$

Find the values of:

- $z_{0.1}$
- $-z_{0.05}$

Normal distribution practice

If $X \sim N(7, 25)$, what is the smallest x among the four choices below such that $P(X > x) = 0.05$?

A. 13.41

B. 15.225

C. 16.8

D. 19.88

Sample Mean and Sample Variance

Suppose we observe a sample of data

$$x_1, x_2, \ldots, x_n.$$

Sample Mean: average value of the data

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Sample Variance: how spread out the data are around the mean

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Sample mean as a random variable

The sample mean changes every time we observe new data.

$\Longrightarrow$ Therefore, $\bar{X}$ is a random variable

$$E[\bar{X}] = \mu$$

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

As the sample size increases:

- The mean stays the same
- The variance shrinks
- The sample mean becomes more stable

Sample mean & variance practice

Let $\bar{X}$ be the mean of a random sample of size 9 from the distribution:

$$f(x) = x/6, \qquad x = 1, 2, 3$$

Find:

- $E[X]$
- $\mathrm{Var}(X)$
- $E[\bar{X}]$
- $\mathrm{Var}(\bar{X})$

Sample mean & variance practice

Chebyshev's Inequality

Suppose a random variable X has mean μ and variance σ^2.

Then for any $k > 0$:

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Interpretation:

- Large deviations from the mean are unlikely
- Higher variance means more spread
- This works for *any* distribution

Law of Large Numbers (LLN)

Suppose:

$$X_1, X_2, \ldots, X_n$$

are i.i.d. observations with mean μ and variance σ^2.

Then:

$$P(|\bar{X} - \mu| \geq \varepsilon) \to 0 \quad \text{as } n \to \infty$$

Interpretation: As sample size grows, the sample mean gets closer to the true mean

Intuition behind the LLN

Suppose the true mean height of UCLA students is 68 inches.

If we take:

- A sample of 2 students, the average could vary a lot
- A sample of 200 students, the average is usually close to 68

Why?

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Larger n means smaller variance for the sample mean.

The Central Limit Theorem (CLT)

Let $\bar{X}$ be the sample mean from a random sample of size n.

If the population has mean μ and variance σ^2, then:

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

is approximately distributed as:

$$N(0, 1)$$

when n is sufficiently large.

Rule of thumb: $n \geq 30$ is often enough.

What does the CLT mean?

Even if the original population is:

- skewed
- discrete
- non-normal

The distribution of the sample mean becomes approximately normal when n is large.

This is one of the most important ideas in statistics.

It allows us to:

- build confidence intervals
- conduct hypothesis tests
- approximate probabilities

Standardizing the sample mean

To apply the CLT, we convert the sample mean into a standard normal variable:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

This measures:

- how far $\bar{X}$ is from μ
- in units of standard deviations

The standard deviation of the mean is:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

CLT Example

Suppose:

- $n = 36$
- $\mu = 15$
- $\sigma = 2$

Approximate:

$$P(14.4 < \bar{X} < 15.6)$$

Standardize:

$$P\left(\frac{14.4 - 15}{2/\sqrt{36}} < Z < \frac{15.6 - 15}{2/\sqrt{36}}\right)$$

$$= P(-1.8 < Z < 1.8)$$

CLT Example Continued

Using the standard normal table:

$$P(-1.8 < Z < 1.8) = \Phi(1.8) - \Phi(-1.8)$$

$$= 0.9641 - 0.0359$$

$$= 0.9282$$

Therefore:

$$P(14.4 < \bar{X} < 15.6) \approx 0.9282$$

Another CLT Practice Problem

A random sample of size $n = 100$ is drawn from a population with:

$$\mu = 30, \qquad \sigma = 3$$

Approximate:

$$P(\bar{X} > 29.5)$$

Key Takeaways

Law of Large Numbers

- The sample mean converges to the population mean

Central Limit Theorem

- The sample mean becomes approximately normal
- Works even when the original population is not normal

Important formulas

$$E[\bar{X}] = \mu, \quad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}, \;\; \forall k > 0$$